

PIKANs: Physics-Informed Kolmogorov-Arnold Networks for Landslide Time-to-Failure Prediction

Jiashan Wan^{a,c}, Na Xia^{a,c}, Ziheng Jian^b, Jinhua Wu^{a,c}, Jingyang Li^a, Kai Wang^c, Mengqi Lian^c

^a*School of Computer and Information Science, Hefei University of Technology, Hefei, 230601, Anhui, China*

^b*School of Robot engineering, Anhui University, Hefei, 230601, China*

^c*Anhui Province Key Laboratory of Industry Safety and Emergency Technology, Hefei University of Technology, Hefei, 230601, Anhui, China*

Abstract

Slope deformation is characterized by pronounced time variability and complexity. Although ground-based synthetic aperture radar (GB-SAR) provides high-frequency, broad monitoring, its strong oscillations and large fluctuations can impair predictive performance. To address this, the raw displacement sequence is first smoothed via misaligned subtraction to suppress high-frequency noise and highlight key deformation trends. A dynamic confidence boundary is then established on the inverse-velocity curve to robustly identify the acceleration start point. Subsequently, physics-informed Kolmogorov-Arnold networks (PIKANs) are formulated to embed the displacement-time evolution equation into the basis-function space of the Kolmogorov-Arnold network (KAN), thereby unifying nonlinear deformation dynamics with the governing physical laws of landslide motion. During model training, an alternating optimization scheme combining Adam and the L-BFGS algorithm accelerates convergence and enhances predictive accuracy. Comparative experiments on field GB-SAR datasets demonstrate that compared with an improved KAN baseline and a physics-informed neural network benchmark, PIKANs reduce the relative error

Email addresses: Corresponding author: 2022030037@mail.hfut.edu.cn (Jiashan Wan), 1033348860@qq.com (Na Xia), w124302075@stu.ahu.edu.cn (Ziheng Jian), jhwu3@iflytek.com (Jinhua Wu), sakajy21@gmail.com (Jingyang Li), wangkai_anhui@foxmail.com (Kai Wang), abzzorro@gmail.com (Mengqi Lian)

in failure-time prediction by 24.24% and 30.23%, respectively. These results confirm that integrating physical equation constraints into neural network parameter updates substantially improves the precision and efficiency of real-time landslide early warning.

Keywords: Failure time of landslides, ground-based synthetic aperture radar, physics-informed neural network, Kolmogorov-Arnold networks

1. Introduction

Early warning systems for landslides in open-pit mines are of critical importance to ensuring safe production. Influenced by factors such as ore deposit conditions, mining technology, and extraction costs, open-pit mining has become the predominant extraction method in many mining operations. However, as the scale of excavation and stripping depth increase, open-pit mine slopes develop free faces, significantly altering the original rock-mass stress distribution and groundwater flow conditions. Under the combined effects of weathering, rainfall erosion, and blasting-induced vibrations, slope rock or coal masses readily undergo deformation and displacement in this new stress environment. Although some slopes may eventually stabilize, others continue to deform or even fail (Wang et al., 2024b). In recent years, several major mine-slope failures, such as the Bingham Canyon Manefay landslide in Utah in May 2013 (Williams et al., 2021), Dexing landslide in Jiangxi Province in May 2018 (Zhou and Ye, 2021), and Xinjing landslide in Alxa League, Inner Mongolia, in February 2023 (Wang et al., 2024b) have resulted in severe casualties and property losses, underscoring the urgency and importance of robust early warning models in mine safety management.

Since the 1960s, early landslide warning has been a focal topic in geological hazard research. Early work relied on empirical formulas and theoretical models; notably, Saito introduced the three creep stages theory of slope deformation and derived the differential equation governing accelerated deformation, thereby laying the theoretical groundwork for subsequent instability-time prediction studies

(Saito, 1965). With the accumulation of long-term monitoring data, statistical and data-driven models based on historical displacement records have gained popularity (Sharifi et al., 2024; Wang et al., 2022). Recent methodological advances include the extraction of deformation probability indexes from displacement data (Zhang et al., 2024), enhanced tangential-angle theory combined with cumulative-acceleration analysis (Ding et al., 2024), and the trend-to-speed ratio model (Du et al., 2023) to improve warning accuracy. Optimization algorithms, such as genetic algorithms (Zhong et al., 2025), Bayesian updating (Zhou et al., 2024b), fuzzy C-means clustering (Wang et al., 2024a), and various inverse-velocity (INV) methods (Zhou et al., 2020; Sharifi et al., 2024), have further advanced the precision and timeliness of landslide-timing predictions. Despite these developments, early warning systems still exhibit limitations (Gupta and Satyam, 2024), particularly their reliance on single deformation indicators (e.g., rate of deformation or cumulative displacement) and predominance of linear modeling, which inadequately capture nonlinear deformation behavior before failure. Thus, integrating multiple warning indicators with nonlinear dynamic modeling is emerging as a key research direction.

Regarding monitoring technology, multi-source data, such as interferometric synthetic aperture radar (InSAR) (Zhou et al., 2024a), global navigation satellite system (GNSS) observations (Lu et al., 2024), light detection and ranging (LiDAR) (An et al., 2024), and unmanned aerial vehicle (UAV) photogrammetry (Zhu et al., 2025), now provide comprehensive support for landslide early warning. Although GNSS offers high-precision, real-time point measurements, its spatial coverage is limited. InSAR and UAV methods deliver aerial deformation maps but due to longer revisit intervals, struggle to capture short-term dynamics. In contrast, ground-based synthetic aperture radar (GB-SAR) combines continuous, high-frequency measurements with non-contact operation, extending from point to area monitoring and thereby, furnishing critical data for landslide risk assessment.

Although GB-SAR offers unique advantages for open-pit mine landslide monitoring, its deformation data remain subject to pronounced oscillations and fluc-

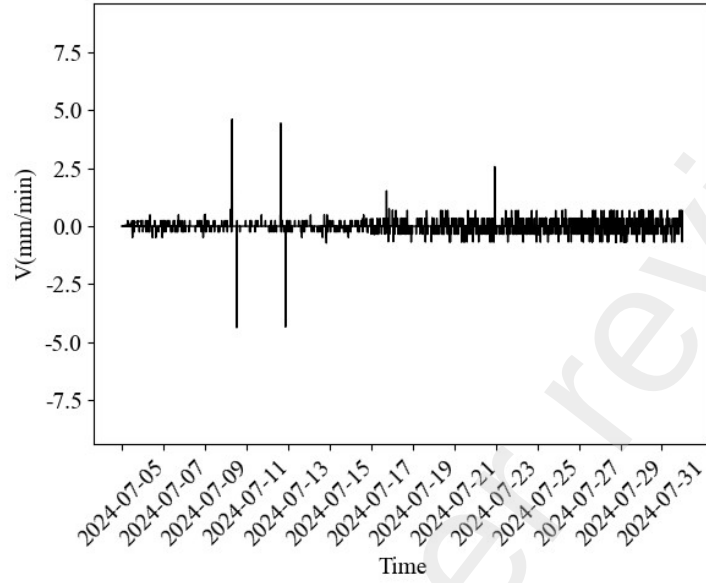


Fig. 1. Velocity-time series curve of GB-SAR monitoring data.

tuations (Hosseiny et al., 2023). Specifically, the raw velocity-time sequence
exhibits significant volatility that obscures the underlying deformation trends
(Izumi et al., 2024). As shown in Fig. 1, the velocity values fluctuate in the
 $[-5 \text{ mm}, 5 \text{ mm}]$ interval, compromising data stability and challenging both the
timeliness and accuracy of early warning.

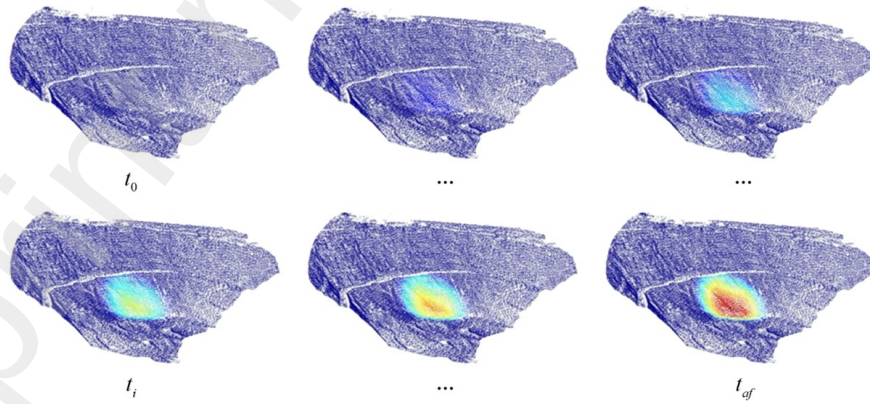


Fig. 2. GB-SAR cumulative-displacement deformation map.

Moreover, a GB-SAR cumulative-displacement map can contain tens to hundreds of thousands of pixels, each associated with its own $S - t$ curve (wherein S denotes displacement). After each scan, displacement data are incrementally updated in a matter of minutes to generate a new displacement map (in Fig. 2, t_0 denotes the start of monitoring and t_{af} represents the actual time of failure). Automatically identifying potential danger zones in these massive, rapidly evolving datasets and accurately predicting the time of instability in real time remains an urgent challenge.

To address this, we propose a GB-SAR-based method for predicting landslide instability time in open-pit mines. We validate its feasibility and accuracy through both numerical simulations and field GB-SAR datasets. The main contributions are:

- Leveraging the Kolmogorov-Arnold theorem, we design physics-informed Kolmogorov-Arnold networks (PIKANs) to replace conventional multi-layer perceptron (MLP) architectures, markedly reducing parameter count and data requirements while enhancing real-time predictive performance.
- We introduce a mechanism that dynamically constructs confidence bounds on the INV curve to detect the starting point (SP) robustly and impose physical parameter constraints when SP is triggered, thereby eliminating the detection lag inherent in traditional onset of acceleration (OOA) methods under noisy conditions.
- We develop a mixed optimization scheme that alternates between the first-order Adam optimizer (for rapid convergence) and the second-order L-BFGS method (for high precision), effectively avoiding local minima and improving both training efficiency and prediction accuracy.

2. Related work

2.1. Displacement-based monitoring techniques

The core of early warning of open-pit mine-slope failure lies in acquiring key information by monitoring surface displacements to predict the precise location, area, and timing of landslides (Wang et al., 2024b). Surface-deformation monitoring techniques are broadly classified into contact (point-based) and non-contact (area-based) methods. Conventional total-station, leveling, and GNSS single-point measurements fall under the contact category, whereas 3D laser scanning and synthetic aperture radar (SAR) constitute typical non-contact, area-based approaches. The measurement modes, accuracies, advantages, and disadvantages of these techniques are summarized in Table I.

SAR technology is categorized into three modes: spaceborne, airborne, and ground-based, according to the platform. Spaceborne and airborne SAR are well suited for monitoring large-area but suffer from substantial orbital positioning errors, long revisit intervals, and slow emergency response. In contrast, GB-SAR offers superior flexibility and real-time capability and has been widely applied to surveillance of mining operations (Ding et al., 2024), bridges (Wang et al., 2024c), and natural slopes (Sestras et al., 2025; Liu et al., 2024a).

Compared to spaceborne and airborne systems, GB-SAR provides shorter observation intervals, higher spatial resolution, greater measurement accuracy, and operational agility, making it particularly suitable for slope-stability monitoring. Specifically, GB-SAR's key advantages include:

- Extensive continuous coverage: GB-SAR can monitor a 3.5 km^2 area with a 2 km range and generate millions of data points, facilitating comprehensive analysis of deformation characteristics and trends.
- Noncontact remote monitoring: GB-SAR performs measurements from several kilometers away without requiring personnel to enter hazardous zones, thereby greatly reducing monitoring risk.

Table. I. Comparison of primary monitoring technologies: advantages and disadvantages

No.	Technique	Measurement mode	Advantages	Disadvantages	Measurement error
1	Traditional surveying instruments (total station, level) (Lizheng et al., 2023; Zhu et al., 2024)	Contact, point-based	Low equipment cost; simple operation	Requires field surveying and manual operation; cannot achieve continuous observation	0.5-5mm
2	GNSS technology (BDS, GPS) (Carlson et al., 2024)	Contact, point-based	Simple operation; real-time dynamic monitoring; all-weather operation	Limited to designated monitoring areas; sensitive to environmental conditions	mm-cm
3	SAR (spaceborne, airborne) (Zhou et al., 2024a; Yan et al., 2024)	Noncontact, area-based	Wide-area coverage; all-weather operation; no ground equipment deployment	Complex data processing; cannot achieve real-time monitoring	mm-m
	SAR (ground-based) (Liu et al., 2024a)	Noncontact, area-based	Wide-area coverage; real-time dynamic monitoring; all-weather operation	Limited to designated monitoring areas	sub-mm
4	3D laser scanning (LiDAR, scanners) (Sestras et al., 2025)	Noncontact, area-based	High spatial resolution; suitable for detailed topographic mapping; rapid large-area scanning	Sensitive to environmental conditions; occlusions or weather changes can affect measurement	mm

- All-time, all-weather real-time operation: By operating in the microwave band, GB-SAR can penetrate rain, fog, and dust, remaining unaffected by adverse weather and enabling continuous real-time monitoring.

Although GNSS technology delivers high accuracy and real-time data, its point-based deployment limits spatial coverage and thus, constrains precise landslide-hazard identification. In contrast, GB-SAR's area deformation monitoring supports large-scale, continuous, and real-time surveillance that captures overall slope deformation trends, overcoming the spatial coverage limitations of GNSS (Carlson et al., 2024).

Nevertheless, GB-SAR data still exhibit strong oscillations and high volatility, which obscure true deformation trends and undermine the reliability of early warning models. Consequently, further analysis and innovation in GB-SAR data processing is urgently needed to enhance mine-slope monitoring performance and strengthen the accurate detection and real-time early warning of landslide risks.

2.2. Landslide early warning methods

Predicting the time of landslide instability is a central task in early warning research, because it underpins the formulation of effective risk mitigation strategies. In recent years, various approaches have been proposed to forecast slope stability and failure timing, which can be broadly categorized into statistical and machine-learning methods and deep-learning methods.

2.2.1. Statistical and machine-learning approaches

Early work on slope stability prediction relied largely on statistical analysis or traditional machine-learning techniques. For example, the artificial bee colony algorithm optimizes ensembles of base classifiers to improve predictive accuracy (Kardani et al., 2021), whereas XGBoost is a gradient-boosting framework that demonstrates strong learning capabilities for handling complex slope datasets (Zhang et al., 2022). A model integrating particle swarm optimization

with support vector regression further enhances slope stability forecasts (Wang et al., 2022).

For failure-time estimation, the INV method is widely adopted due to its simplicity and intuitive formulation: By relating the reciprocal of deformation velocity to time, it infers the point of failure. However, traditional INV performs poorly outside the accelerated creep phase, and its accuracy diminishes when deformation rates decelerate. To address these limitations, numerous INV variants such as the M-INV (Zhou et al., 2020), T-INV (Sharifi et al., 2024), and INSRA (Zhou and Ye, 2021) have been developed, demonstrating improved performance in both acceleration and deceleration stages when validated against monitoring data.

Although these statistical and machine-learning techniques offer feasible avenues for slope-stability assessment, they struggle to capture the inherently nonlinear dynamics of landslide processes, particularly during rapid acceleration phases (Zhou et al., 2024b). Consequently, there is an urgent need for higher-order modeling frameworks that can enhance both the precision and robustness of instability-time predictions.

2.2.2. Deep-learning approaches

With the expansion of data availability and computational power, deep-learning models have been increasingly applied to slope stability and landslide forecasting (Nava et al., 2023). Unlike traditional statistical or shallow-learning algorithms, deep models can perform flexible nonlinear mappings in high-dimensional feature spaces, offering superior feature extraction and representation capabilities (Thirugnanam, 2023).

Among these, Kolmogorov-Arnold networks (KANs) leverage the Kolmogorov-Arnold theorem to address complex nonlinear problems effectively (Somvanshi et al., 2024; Vaca-Rubio et al., 2024). For instance, the TKAN model combines additive operations with multilayer mapping to enhance robustness and scalability in high-dimensional, nonlinear system modeling (Genet and Inzirillo, 2024). TimeKAN employs multi-frequency decomposition to segment time series, fol-

lowed by frequency mixing, thereby capturing multi-scale nonlinear features in landslide prediction (Huang et al., 2025).

In contrast, physics-informed neural networks (PINNs) integrate differential equations or mechanical constraints directly into the training objective, fusing data-driven learning with physical mechanisms (Torres and Niepert; Toscano et al., 2024). PINNs have been adapted for landslide forecasting (Liu et al., 2024c; Dahal and Lombardo, 2025), capitalizing on their ability to embody geomechanical laws and improve predictive accuracy. Several enhanced PINN variants have emerged: RoPINN extends point-set optimization to regional domains via a region-optimized strategy, boosting training efficiency and generalization (Wu et al., 2025); VW-PINNs address the large number of hyperparameters in original PINNs to strengthen generalizability (Song et al., 2025); and SVD-PINNs (Gao et al., 2022) and low-rank PINNs (Cho et al., 2023) reduce model complexity and resource demands while maintaining accuracy.

In summary, KANs excel at modeling nonlinear structures and multi-dimensional couplings, whereas PINNs enhance interpretability and applicability through physical constraints. Given the high-frequency oscillations and volatility inherent in GB-SAR data, an integrated framework (e.g., embedding PINN constraints in a KAN architecture) is proposed to suppress noise effects during critical acceleration phases. This hybrid methodology addresses the limitations of current approaches and enables more accurate and reliable predictions of landslide instability timing.

3. Problem denition

Landslide events typically progress through a three-stage evolution from initial deformation to eventual failure: the initial deformation stage, the constant-rate deformation stage, and the accelerating deformation stage (FUKUZONO, 1985). The deformation characteristics in each stage differ markedly, as shown in Fig. 3(a). During the initial deformation stage (segment AB in the figure), the slope undergoes decelerating deformation: Displacement acceleration gradually

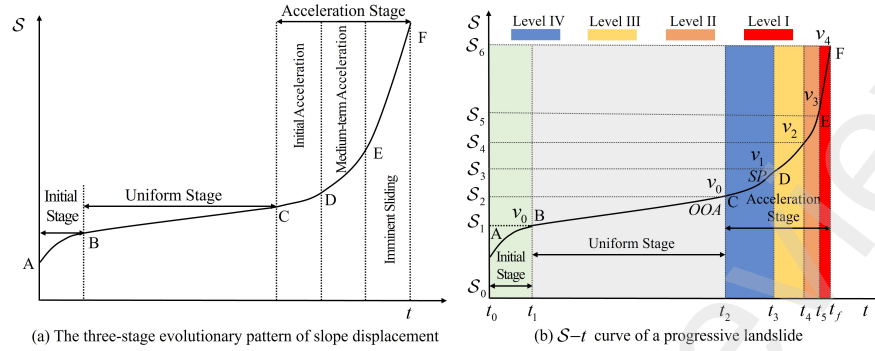


Fig. 3. Three-stage evolution of slope displacement, with the acceleration stage divided into four warning levels.

decreases, and both the slope of the displacement-time curve and the tangent angle of the displacement decline continually. In the constant-rate deformation stage (segment BC), displacement and time exhibit a strong positive correlation, the curve's slope approaches a constant value, deformation velocity fluctuates around a stable mean, and acceleration remains near zero. In the accelerating deformation stage (segment CF), the positive correlation between displacement and time weakens, the correlation coefficient decreases, and both deformation velocity and acceleration increase, with the tangent angle of the displacement curve rising progressively; see Fig. 3(b). Research has indicated that the rate and acceleration characteristics immediately before and after the onset of the accelerating deformation stage have a significant impact on the variation of the tangent angle of the displacement-time curve (Qiang et al., 2009; Liu et al., 2024b).

For open-pit mine slopes that have already undergone failure, the acceleration phase of a landslide can be identified by analyzing the displacement-time (e.g., $S - t$) curve recorded by GB-SAR at the failure event. However, raw GB-SAR monitoring data often exhibit substantial volatility and oscillations, which obscure the true deformation trend and consequently undermine the effectiveness of early warning. In practical monitoring, these fluctuations may conceal the genuine deformation behavior, making it difficult to assess intu-

220 itively whether the slope is approaching instability and thus, diminishing warn-
 221 ing accuracy.

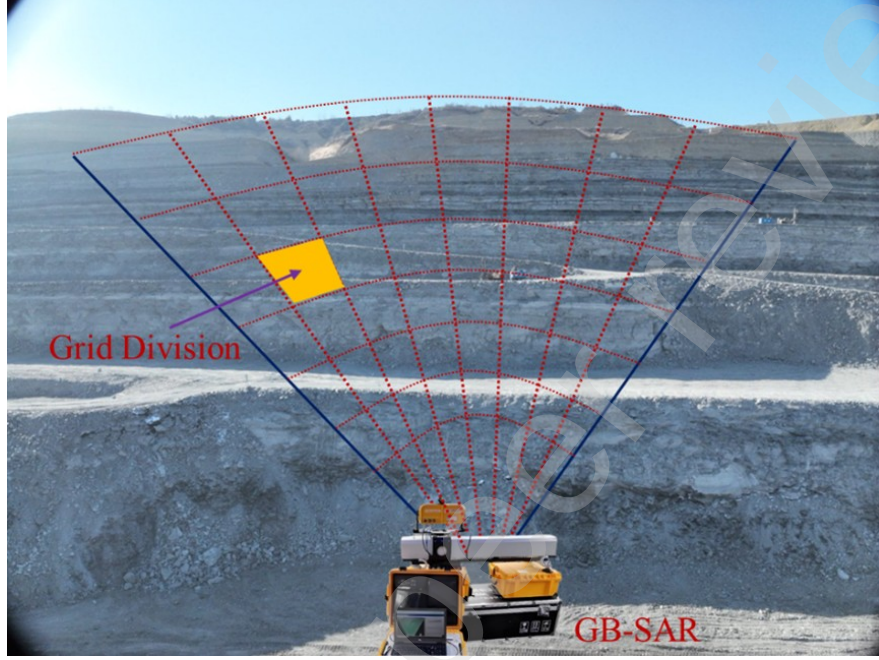


Fig. 4. Schematic diagram of the GB-SAR monitoring setup for an open-pit mine slope.

222 **Grid partitioning.** In the preprocessing of raw GB-SAR data, a grid-
 223 based aggregation method is employed to spatially consolidate area-based mea-
 224 surements. Specifically, the monitored region is divided into a regular array of
 225 grid cells (see Fig. 4), each of which contains n measurement points. Let s_i
 226 denote the displacement recorded at the i^{th} point in a given cell; then the cell's
 227 aggregate displacement S can be computed as:

$$\mathcal{S} = \frac{1}{n} \sum_{i=1}^n s_i \quad (1)$$

228 Grid cell size selection is dictated by the radar's spatial resolution to guaran-
 229 tee the detection of fine-scale deformation features. Both the spatial resolution
 230 of the displacement dataset and the reliability of subsequent deformation anal-
 231 yses are contingent on the precision of grid partitioning.

Time-series processing. To ensure consistency in deformation-velocity calculations and standardize velocity units $\{(t_i, v_i)\}_{i=1}^N$, the lagged-differencing method is applied to the cumulative displacement time series. By selecting a fixed time interval t_i , the deformation velocity v_i over that interval is computed as:

$$\begin{cases} \mathcal{S}_{t_i} = \mathcal{S}_{t_0+t_i} - \mathcal{S}_{t_0} \\ v_i = \frac{\mathcal{S}_{t_i}}{t_i} \end{cases} \quad (2)$$

where $\mathcal{S}_{t_0}$ denotes the cumulative displacement at time t_0 , $\mathcal{S}_{t_0+t_i}$ is the displacement at time $t_0 + t_i$, and v_i is the deformation velocity over the interval $[t_0, t_0 + t_i]$.

4. Proposed method: PIKANs

The overall workflow of the proposed GB-SAR-based instability time prediction method for open-pit mine landslides is illustrated in Fig. 5. First, raw GB-SAR monitoring data are preprocessed and the SP is dynamically determined using a normal confidence interval criterion. Next, to capture the nonlinear dynamics and complex physical laws of landslide motion fully, we develop the PIKANs model. Built in a PINN framework, PIKANs embed both the INV method and KANs to enhance prediction accuracy and real-time performance.

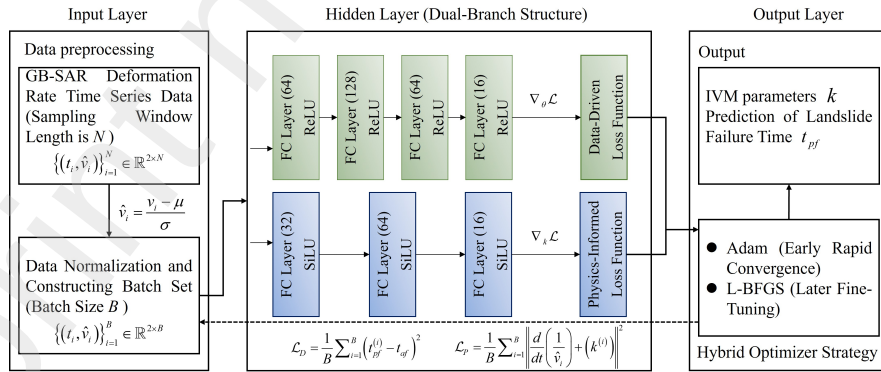


Fig. 5. Network architecture of the PIKANs method.

248 4.1. Dynamic identification method for the starting point

249 Measurement errors are inevitable during the constant-rate deformation stage
 250 of a slope. Under the assumption of normally distributed errors, the measure-
 251 ment noise inherent in GB-SAR monitoring can be treated as a random vari-
 252 able following a normal distribution. Hence, during the constant-rate stage,
 253 the observed deformation-velocity samples—each contaminated by measure-
 254 ment error—also conform to a normal distribution. The probability density
 255 function of the normal distribution is given by:

$$f(v) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{(v - \mu)^2}{2\sigma^2} \right] \quad (3)$$

256 where μ denotes the mean and σ is the standard deviation of the velocity sam-
 257 ples.

258 Skewness is a statistic that quantifies the asymmetry of a distribution. The
 259 skewness γ of the velocity sample set $\{(t_i, v_i)\}_{i=1}^N$ is defined as:

$$\gamma = \frac{\frac{1}{N-1} \sum_{i=1}^N (v_i - \mu)^3}{\sigma^3} \quad (4)$$

260 where n is the number of samples. A skewness of $\gamma = 0$ indicates perfect
 261 symmetry (consistent with a normal distribution); $\gamma > 0$ signifies right-skewed
 262 data; and $\gamma < 0$ indicates a left-skewed distribution.

263 Assuming the slope is in the constant-rate deformation stage, the mea-
 264 sured deformation-velocity random variable v_n follows a normal distribution:
 265 $v_n \sim \mathcal{N}(\mu, \sigma^2)$. During this stage, as monitoring data are continuously updated,
 266 new velocity samples $(v_{n+1}, v_{n+2}, \dots)$ are generated at each time step. While
 267 repeatedly testing the updated samples against the previous normal-distribution
 268 confidence interval, if at some time a new observation v_n falls outside that inter-
 269 val, and subsequent observations persistently remain outside this interval while
 270 exhibiting positive sample skewness, the velocity can no longer be considered
 271 normally distributed. The time at which this first occurs is taken as the INV
 272 method's analysis SP.

273 The random variable v is therefore modeled as $v \sim \mathcal{N}(\mu, \sigma^2)$, where the
 274 mean μ and standard deviation σ are computed from the most recent n samples
 275 by:

$$\begin{cases} \mu = \frac{\sum_{i=1}^n v_i}{n} \\ \sigma = \frac{\sum_{i=1}^n (v_i - \bar{v})^2}{n-1} \end{cases} \quad (5)$$

276 According to the INV method, landslide instability-time prediction then com-
 277 mences from this identified SP. Fixing this analysis origin and examining the
 278 subsequent trend in velocity data enables forecasting both the likelihood and
 279 approximate timing of slope failure.

280 4.2. Failure time of landslide prediction model

281 To enhance the robustness and accuracy of failure-time predictions, we pro-
 282 pose the PIKANs method. The central concept of PIKANs is to embed the
 283 conventional INV approach as a physical constraint in a neural network—
 284 specifically, KANs—by jointly minimizing the data-fitting loss and a physics-
 285 consistency loss. This hybrid optimization enables high-precision forecasting of
 286 landslide instability time. Beyond improving the model’s capacity to capture
 287 the dynamic behavior of landslides, PIKANs also bolster the robustness and
 288 reliability of predictions. Fig. 5 illustrates the PIKANs architecture.

289 4.2.1. Mathematical formulation of the physical constraint

290 The traditional INV method assumes a linear relationship between the re-
 291 ciprocal of deformation rate and time, which can be expressed as:

$$\frac{1}{v} = [k(a-1)]^{\frac{1}{a-1}} (t_{pf} - t)^{\frac{1}{a-1}} \quad (6)$$

292 where v is the deformation rate, t_{pf} is the predicted failure time, and k is the
 293 rate-decay coefficient (a physical parameter).

294 Extensive empirical studies (FUKUZONO, 1985; Zhou and Ye, 2021) have
 295 indicated that a typically lies between 1.5 and 2.2, with approximately 50% of
 296 measured cases yielding $a = 2$. This observation makes the linear fit of INV

297 versus time a simple yet effective landslide-timing predictor. Consequently, Eq. 6
 298 can be simplified to:

$$\frac{1}{v} = k(t_{pf} - t) \quad (7)$$

299 To capture nonlinear deformation features better, we reformulate the conven-
 300 tional INV relation as a differential equation:

$$\frac{d}{dt} \left(\frac{1}{v} \right) = -k \quad (8)$$

301 Whereas the classic INV method depends on linear assumptions, PIKANs
 302 leverages a neural network architecture to learn nonlinear behaviors (e.g., sud-
 303 den transitions and oscillatory patterns) observed in geologically complex slope
 304 systems. By embedding the INV differential-equation constraint into the net-
 305 work's loss function, PIKANs combines physical interpretability with nonlinear
 306 fitting capacity, yielding accurate predictions of landslide instability time and
 307 providing a robust quantitative basis for early warning.

308 4.2.2. Dual-branch neural network architecture

309 PIKANs employs a dual-branch, fully-connected neural network architec-
 310 ture, as illustrated in Fig. 5. To facilitate efficient training, the input layer ac-
 311 cepts preprocessed GB-SAR monitoring data, which are normalized to $\hat{v}_i = \frac{v_i - \mu}{\sigma}$
 312 and partitioned into mini-batches $\{(t_i, \hat{v}_i)\}_{i=1}^B$ of size B .

313 The hidden layers split into two functional branches: one for predicting the
 314 landslide failure time t_{pf} , and the other for estimating the physical parameter
 315 k . The t_{pf} prediction branch consists of four fully connected layers with ReLU
 316 activations, enhancing the network's ability to extract nonlinear features. The k
 317 estimation branch features three fully connected layers using SiLU activations,
 318 which improve expressive capacity and stabilize training.

319 Finally, the output layer fuses both branches, yielding simultaneous predic-
 320 tions of the failure time t_{pf} and the physical parameter k . This dual-output
 321 design enhances both the accuracy and robustness of landslide-time forecasts.

322 4.2.3. Kolmogorov-Arnold network layer

323 To capture the nonlinear features of landslide motion more effectively, we
 324 leverage the Kolmogorov-Arnold theorem by defining each basis function $\phi_{j,i}(\cdot)$
 325 as a univariate nonlinear transform that incorporates the physical parameter k
 326 as an additional input term:

$$\phi_{j,i}(\hat{v}_i, k) = \psi_{j,i}(\omega_{j,i} \cdot \hat{v}_i + b_{j,i} \cdot k) \quad (9)$$

327 where $\psi_{j,i}(\cdot)$ is a learnable spline function and $\omega_{j,i}$ and $b_{j,i}$ are the layer's weight
 328 and bias parameters; k is updated in real time via the INV method.

329 The output of the KAN layer is then formed by an additive fusion of the
 330 data-driven mode and a physics-informed mode:

$$f_{KAN}(X, k) = \sum_{j=1}^{2n+1} \Phi_j \left(\sum_{i=1}^n \phi_{j,i}(\hat{v}_i, k) \right) \quad (10)$$

331 where $\Phi_j(\cdot)$ is a higher-order combination function that is dynamically coupled
 332 with k through parameter sharing.

333 The input to the first layer, h_0 is the normalized rate-time sequence $X =$
 334 $\{(t_i, \hat{v}_i)\}_{i=1}^B$ from GB-SAR. Stacking L such as KAN layers yields the final
 335 failure-time predictor as a composition of these nonlinear transformations:

$$t_{pf} = KAN(h) = (\Phi_L \circ \dots \circ \Phi_\ell \circ \dots \circ \Phi_1) h \quad (11)$$

336 where $\circ$ denotes function composition (i.e., the output of layer $\ell - 1$ serves as the
 337 input to layer ℓ). Each layer's nonlinear transformation progressively extracts
 338 complex patterns from the input, and the final output of t_{pf} is the predicted
 339 landslide instability time.

340 This layer-by-layer composition illustrates how KANs process and transform
 341 input data at each stage, integrating physical constraints and nonlinear modeling
 342 to provide accurate and robust predictions of slope failure timing.

343 4.2.4. Loss function

344 The total loss for PIKANs is formulated as a weighted sum of the data-fitting
 345 term and the physics-constraint term:

$$\mathcal{L} = \mathcal{L}_D + \lambda \mathcal{L}_P \quad (12)$$

where $\lambda \in [0, 1]$ is a hyperparameter chosen via validation.

Data loss term $\mathcal{L}_D$. If a ground-truth failure time t_{af} is available, we measure the prediction error with the mean squared error:

$$\mathcal{L}_D = \frac{1}{B} \sum_{i=1}^B \left(t_{pf}^{(i)} - t_{af} \right)^2 \quad (13)$$

where $t_{pf}^{(i)}$ and $t_{af}^{(i)}$ denote the predicted and true instability times for sample i .

Physics-constraint term $\mathcal{L}_P$. We enforce the INV differential equation by penalizing its residual:

$$\mathcal{L}_P = \frac{1}{B} \sum_{i=1}^B \left\| \frac{d}{dt} \left(\frac{1}{\hat{v}_i} \right) + \left(k^{(i)} \right) \right\|^2 \quad (14)$$

where the time derivative $\frac{d}{dt} \left(\frac{1}{\hat{v}_i} \right)$ is computed via automatic differentiation to avoid numerical-differentiation errors and $k \in \mathbb{R}^+$ is constrained to be positive by applying a Softplus activation.

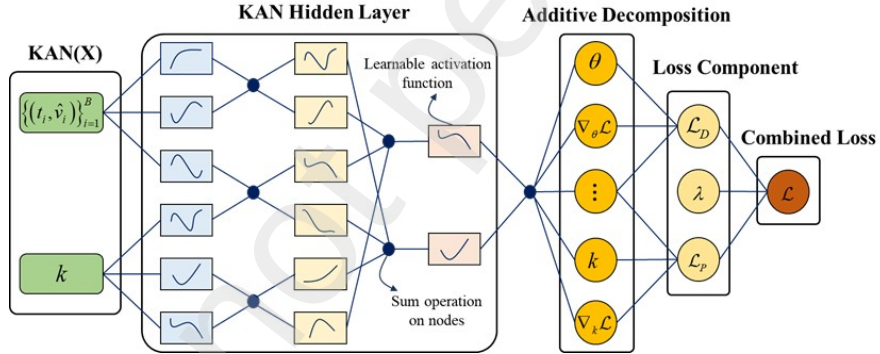


Fig. 6. PIKANs hybrid optimization strategy.

4.2.5. Hybrid optimization strategy

The network is optimized in two phases. To enhance training efficiency and convergence precision, PIKANs employs a phased hybrid optimization strategy: First, when the model operates in a high-gradient regime, a first-order optimizer (e.g., Adam) is used to update the additive decomposition layers rapidly, enabling the network to track real-time GB-SAR data updates and dynamically adjust its parameters. Once the physical-constraint residual

Algorithm 1: PIKANs training procedure

Input: Deformation velocity sequence $\{(t_i, v_i)\}_{i=1}^N$; Physical constraint weight λ ; Maximum training epochs $T_{\max}$; Convergence threshold ε ; learning rate η .

Output: Predicted failure time t_{pf} ; Physical INV parameter k .

2 Data Preprocessing: Normalize $\hat{v}_i = (v_i - \mu)/\sigma$, and form a dataset X of batch size B .

3 Initialization: KAN network $\mathcal{N}_\theta$ with random weights θ ; Physical parameter $k \sim \mathcal{U}(0, 1)$; Adam optimizer $\mathcal{O}_A(\theta, k, \eta)$ and L-BFGS optimizer $\mathcal{O}_B$.

4 for epoch = 1 **to** $T_{\max}$ **do**

 Randomly sample B examples $\{(t_i, \hat{v}_i)\}_{i=1}^B \sim X$;

foreach $i = 1, \dots, B$ **do**

$(t_{\text{pf}}^{(i)}, \tilde{k}^{(i)}) \leftarrow \mathcal{N}_\theta(t_i, \hat{v}_i)$;

$k^{(i)} \leftarrow \text{Softplus}(\tilde{k}^{(i)})$ // Non-negative constraint

end

if the actual failure time t_{af} is available **then**

 Compute data loss $\mathcal{L}_D \leftarrow \frac{1}{B} \sum_{i=1}^B (t_{\text{pf}}^{(i)} - t_{af})^2$ // Eq. 13

end

else

$\mathcal{L}_D \leftarrow \lambda \mathcal{L}_P$

end

 Compute physical constraint loss

$$\mathcal{L}_P \leftarrow \frac{1}{B} \sum_{i=1}^B \left\| \frac{d}{dt} \left(\frac{1}{\hat{v}_i} \right) + k^{(i)} \right\|^2 \quad (\text{Eq. 14}).$$

 Compute total loss $\mathcal{L} \leftarrow \mathcal{L}_D + \lambda \mathcal{L}_P$ (Eq. 12).

 Compute gradients $\nabla_\theta \mathcal{L}$ and $\nabla_k \mathcal{L}$;

if $\|\nabla_\theta \mathcal{L}\|_2 > 10^{-3}$ **then**

 Update via Adam: $(\theta, k) \leftarrow \mathcal{O}_A(\theta, k, \nabla \mathcal{L})$;

else

 Update via L-BFGS: $(\theta, k) \leftarrow \mathcal{O}_B(\theta, k, \mathcal{L})$;

end

if $\mathcal{L} < \varepsilon$ **then**

break // Early stopping

end

if epoch mod 10 = 0 **then**

 Evaluate RE on the test set // Eq. 16

end

27 end

28 Output:

$$t_{\text{pf}} = \frac{1}{N} \sum_{i=1}^N t_{\text{pf}}^{(i)}, \quad k = \frac{1}{N} \sum_{i=1}^N k^{(i)}.$$

return t_{pf}, k .

falls below a predefined threshold, a second-order method (e.g., L-BFGS) is introduced to finely tune regions with larger physical errors, thereby effectively

avoiding local minima (see Algorithm 1). By organically integrating data-driven learning with physical priors, this strategy not only markedly improves generalization under noisy conditions but also achieves an optimal balance between training efficiency and predictive accuracy in complex physical scenarios.

Dynamic parameter optimization. Leveraging the statistical properties of each real-time GB-SAR data batch $\{(t_i, \hat{v}_i)\}_{i=1}^B$, the model first determines whether the landslide motion has entered a new acceleration phase. When the SP indicates an inflection, the dynamic parameter optimization module adaptively updates or reinitializes the physical parameter k and adjusts its allowable range based on the SP result. Specifically, the updated bounds are set as:

$$k_{\min} = \bar{k} - 3\sigma_k \leq k \leq k_{\max} = \bar{k} + 3\sigma_k \quad (15)$$

where $\bar{k}$ and σ_k denote the historical mean and standard deviation of k , respectively.

This mechanism ensures that even under strong noise perturbations, k remains constrained to a physically plausible interval, preventing sudden noise-induced shifts and thereby continuously and accurately capturing the true acceleration process of the landslide.

5. Cases studies

5.1. Overview of the open-pit mines

5.1.1. Slope monitoring data

The Dexing open-pit copper mine is in Dexing City, Jiangxi Province, China ($117^\circ 37' 00'' - 117^\circ 48' 00''$ E, $28^\circ 59' 00'' - 29^\circ 05' 00''$ N). The mining area measures 6.6km east-west by 2.2km north-south, with a maximum depth of 420m . Bench heights range from 9m to 32m , and overall pit-slope angles vary between 45 and 75 . On May 5, 2018, a local landslide occurred on the north slope, with an estimated volume of approximately $1.5 \times 10^4\text{m}^3$. During the event, GB-SAR was deployed for high-precision monitoring, yielding extensive

displacement records. Data were acquired at 7-minute intervals from 14:24 on May 4, 2018, to 14:54 on May 5, 2018, corresponding to 210 sampling epochs.

The Xinjing open-pit coal mine is Alxa League, Inner Mongolia ($105^{\circ} 38' 43'' - 105^{\circ} 40' 03''$ E, $37^{\circ} 58' 09'' - 37^{\circ} 58' 49''$ N). On February 22, 2023, two successive landslides occurred, with a combined volume of approximately $7.56 \times 10^6 m^3$. Prefailure indicators such as surface cracking, sporadic rockfalls, and localized collapses were observed; however, the absence of a monitoring and early warning system precluded advance alerts. To reconstruct the deformation history, Sentinel-1 high-resolution SAR images covering January 2021 to February 2023 were retrieved via the Google Earth Engine platform.

5.1.2. Experiment settings

All numerical simulations and model training were conducted on a high-performance computing workstation running Windows 10. The computational environment featured an Intel Core i9-13900K CPU (5.2 GHz), an NVIDIA GeForce RTX 4090 GPU, and software frameworks including Python 3.8, FLAC3D V7.0, and PyTorch. Sentinel-1 SAR data served as the remote-sensing input, with preprocessing and batch analyses performed in Google Earth Engine. Experimental parameters and hyperparameters were selected in accordance with established studies to ensure reproducibility and reliability of results.

5.1.3. Evaluation metrics

To evaluate the accuracy of the proposed model, we employ the relative error (RE), defined as the ratio of the absolute difference between the predicted failure time and actual failure time to the total sampling duration. Formally:

$$RE = \frac{|t_{pf} - t_{af}|}{t_{total}} \times 100\% \quad (16)$$

where t_{pf} is the predicted failure time, t_{af} is the actual landslide occurrence time, and t_{total} is the sampling period of the monitoring data. A smaller RE indicates higher predictive accuracy.

5.1.4. Comparative methods

We benchmark our method against several state-of-the-art landslide-instability time prediction techniques: Verhulst (Liu et al., 2024b), GM(1,1) (Wu et al., 2020), INSRA(Zhou and Ye, 2021), (Zhou et al., 2020), TimesNet (Wu et al., 2023), TIMEKAN (Huang et al., 2025), and PINN (Cui et al., 2024). These models represent the prevailing approaches in landslide time forecasting. By comparing PIKANs against them, we demonstrate the advantages of our framework in accuracy and robustness.

5.2. Open-pit mine case study

5.2.1. Dexing open-pit mine landslide

We applied multiple predictive models to the Dexing landslide dataset and found that the proposed PIKANs method forecasted the failure time more accurately and timely than traditional approaches. As shown in Fig. 7, the slope transitioned from the initial deformation stage to the constant-rate stage at 19:56 on May 4, 2018; the shift from the constant-rate to the acceleration stage occurred at 07:05 on May 5, 2018, which was identified as the OOA point.

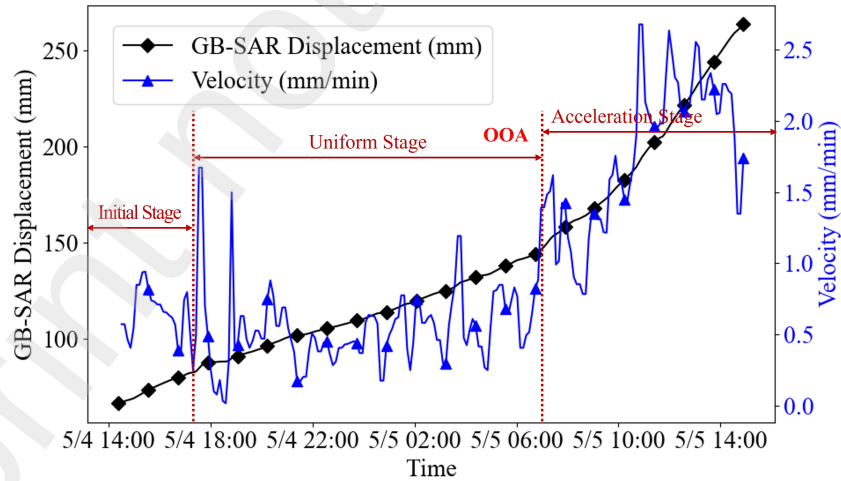


Fig. 7. GB-SAR-monitored landslide displacement and velocity trends at the Dexing open-pit mine.

Using displacement data from the OOA point onward, we estimated the landslide-instability time. The SP was detected via the normal-distribution confidence interval $[\mu - 1.96\sigma, \mu + 1.96\sigma]$: It is defined as the first instance when the velocity data's skewness becomes positive and subsequent samples fall outside the confidence bounds; see Fig. 8(a). This procedure located the SP at 10:35 on May 5, 2018, corresponding to 207.65 monitoring cycles (averaged over the last 10 measurements); see Fig. 8(b). Given the GB-SAR sampling interval of 7 minutes, the model predicted the landslide would occur at approximately 14:38 on May 5, 2018 (i.e., between the GB-SAR scans at 14:33 and 14:40). The actual failure took place at 14:54 on May 5, 2018; our predicted time t_{pf} preceded the true failure time t_{af} by 16 minutes, confirming the strong predictive performance of the PIKANs framework.

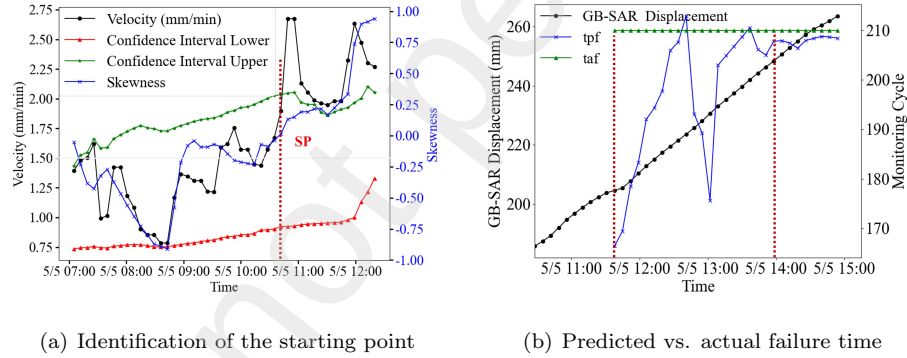


Fig. 8. Dexing open-pit mine landslide: starting point identification and instability-time prediction.

Based on the results shown in Table II, PIKANs exhibited clear advantages in the landslide-instability time prediction task, as detailed below:

Versus traditional statistical methods: Conventional statistical analyses or INV methods rely on linear assumptions or simple extrapolation and thus, failed to capture the nonlinear dynamics of the acceleration phase, resulting in markedly lower predictive accuracy than PIKANs.

Table. II. Landslide failure time prediction for Dexing open-pit mine using different methods.

Category	Method	Predicted time t_{pf}	True failure time t_{af}	RE
Traditional statistical methods	Verhulst	2018-05-05 11:47	2018-05-05 14:54	12.72%
	GM(1,1)	2018-05-05 12:16		10.75%
	INSRA	2018-05-05 13:44		5.24%
	M-INV	2018-05-05 14:07		3.20%
Deep-learning methods	TimesNet	2018-05-05 14:16	2018-05-05 14:54	3.06%
	TimeKAN	2018-05-05 14:28		1.77%
Physics-informed methods	PINN	2018-05-05 14:34	2018-05-05 14:54	1.37%
	PIKANs	2018-05-05 14:38		1.09%

Versus deep-learning methods: Compared with TimesNet and TimeKAN, PIKANs reduced the RE by 64.38% and 38.42%, respectively. Purely data-driven models, lacking physical constraints and operating on limited datasets, cannot effectively model the complex physical mechanisms of landslide motion.

Versus physics-informed methods: Relative to PINNs, PIKANs further lowered the RE by 20.44%. This gain stems from two key enhancements: (a) the additive-decomposition structure of KANs, which explicitly separates data features from physical laws, and (b) the hybrid Adam and L-BFGS optimization strategy, which boosts training efficiency and model precision.

Moreover, whereas other benchmark methods generally predicted failure times earlier than the actual event, PIKANs, which leverage GB-SAR's high-accuracy, high-frequency displacement data, delivered forecasts that were both more accurate and more robust in predicting the time to instability.

5.2.2. Xinjing open-pit mine landslide

At 13:12 on February 22, 2023, a severe landslide occurred at the Xinjing open-pit coal mine in Alxa Left Banner, Inner Mongolia. We analyzed high-resolution Sentinel-1 imagery from January 2021 through February 2023 via

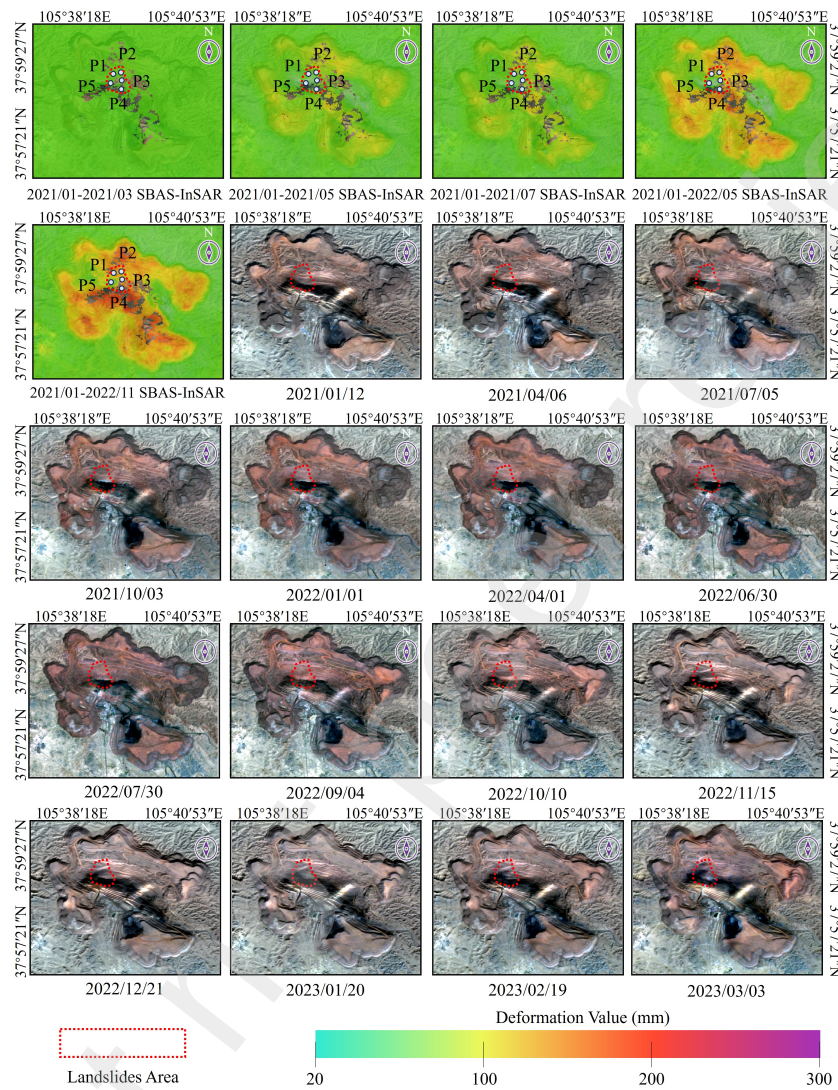


Fig. 9. Deformation analysis based on high-resolution Sentinel-1 imagery from January 2021 to March 2023.

the Google Earth Engine platform. Monitoring results reveal an anomalous deformation area of approximately 1km^2 on the mine slope, with a maximum cumulative displacement of about 290mm (Fig. 9). Prior to failure, the study area underwent three distinct acceleration phases, whose deformation charac-

470 teristics closely correlated with changes in slope angle and excavation depth
 471 (Fig. 10):

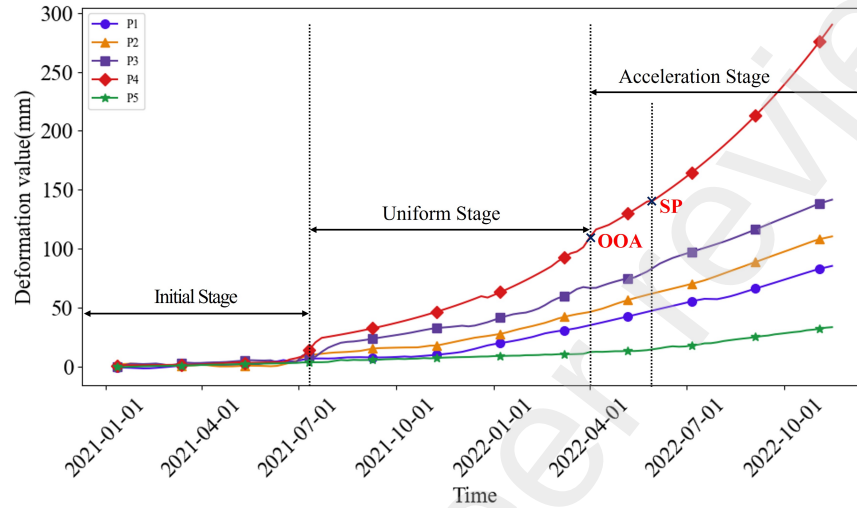


Fig. 10. Cumulative deformation curves at various monitoring points from January 2022 to November 2022.

472 **Phase I (January 2021-late July 2021):** Excavation height and slope
 473 angle were relatively low; cumulative deformation remained under 20 mm, in-
 474 dicating overall slope stability.

475 **Phase II (late July 2021-April 2022):** As excavation depth and an-
 476 gle increased, deformation intensified, reaching approximately 84.87mm. The
 477 deformation center migrated gradually down-slope, highlighting the impact of
 478 excavation activities and a decline in stability.

479 **Phase III (April 2022-February 2023):** Further increases in slope angle
 480 and excavation depth accelerated deformation, with cumulative displacement
 481 surpassing 180.37mm. The deformation center expanded across the entire slope
 482 toe and reached its maximum over 7 months, signaling proximity to the critical
 483 failure state.

484 When performing SBAS-InSAR(Yan et al., 2024) inversion with Sentinel-1
 485 data to construct displacement time series, various landslide-timing prediction

486 methods can be applied; however, their forecasts are generally inferior to those
 487 based on GB-SAR data and frequently predict the event later than its actual
 488 occurrence (see Table III). This systematic bias arises primarily because the
 489 Xinjing mine landslide dataset relies on monthly cumulative displacement rather
 490 than daily measurements, resulting in lower temporal resolution and reduced
 491 predictive accuracy.

492 In contrast, our proposed method achieves earlier and more accurate predic-
 493 tions for two main reasons. First, by dynamically adjusting the SP, we capture
 494 the OOA with high precision. Second, we continuously update and correct the
 495 PIKANs model parameters in real time, enabling the network to reflect the land-
 496 slide's evolving dynamics, especially during the acceleration phase. Even when
 497 applied to lower-frequency remote-sensing data, our approach effectively over-
 498 comes the lag inherent in conventional methods, thereby significantly improving
 499 both the timeliness and the accuracy of landslide-time forecasts.

Table. III. Landslide failure time prediction for Xinjing open-pit mine using different methods.

Category	Method	Predicted time t_{pf}	True failure time t_{af}	RE
Traditional statistical methods	Verhulst	2022-10-22	2023-02-22	18.58%
	GM(1,1)	2023-05-06		14.16%
	INSRA	2022-11-09		15.93%
	M-INV	2022-12-03		12.39%
Deep-learning methods	TimesNet	2022-12-21	2023-02-22	9.73%
	TimeKAN	2022-12-27		8.85%
Physics-informed methods	PINN	2022-11-27	2023-02-22	13.27%
	PIKANs	2023-01-02		7.96%

500 According to the results in Table III, the predictive performance of all meth-
 501 ods deteriorated, highlighting the indispensable role of high-frequency monitor-
 502 ing techniques, such as GB-SAR, in improving both the accuracy and timeli-
 503 ness of landslide forecasting. Unlike the findings in Table II, the INV method

and its variants show no marked difference in RE compared to other classical approaches, indicating that their performance is highly dependent on the data-acquisition frequency. Consequently, in scenarios using high-frequency GB-SAR data, embedding the INV method as a physical-constraint term in PIKANs is not only feasible but also essential to enhance model performance.

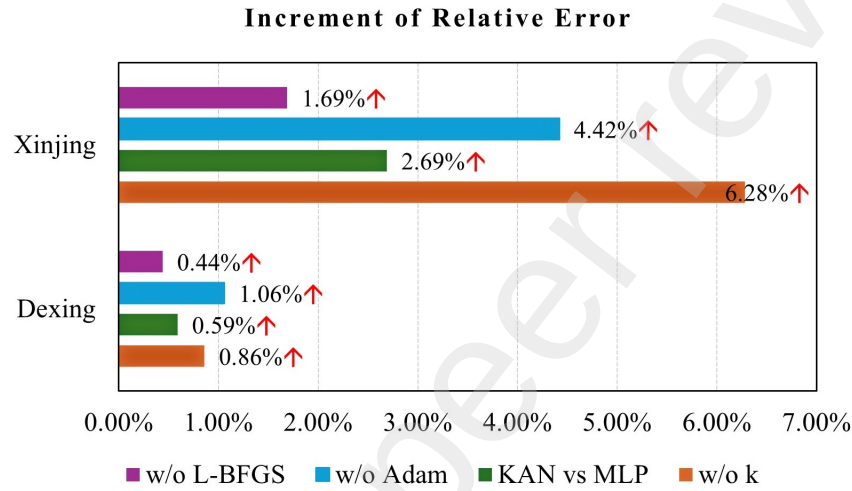


Fig. 11. Relative error performance under ablation of corresponding components on two datasets.

5.2.3. Ablation study

As shown in Fig. 11, ablation experiments were conducted to quantify the contribution of key PIKANs components to overall performance:

(1) **Without physical constraint k** : By removing the physical-constraint branch and retaining only the KAN branch for landslide-time prediction, the RE increases by 79%, demonstrating that the physical constraint is critical for both model accuracy and interpretability.

(2) **KAN vs MLP**: Substituting the original KAN layers with a conventional MLP yields a 54% increase in RE, indicating that KANs substantially outperform MLPs in capturing complex nonlinear dynamics and physics-data couplings.

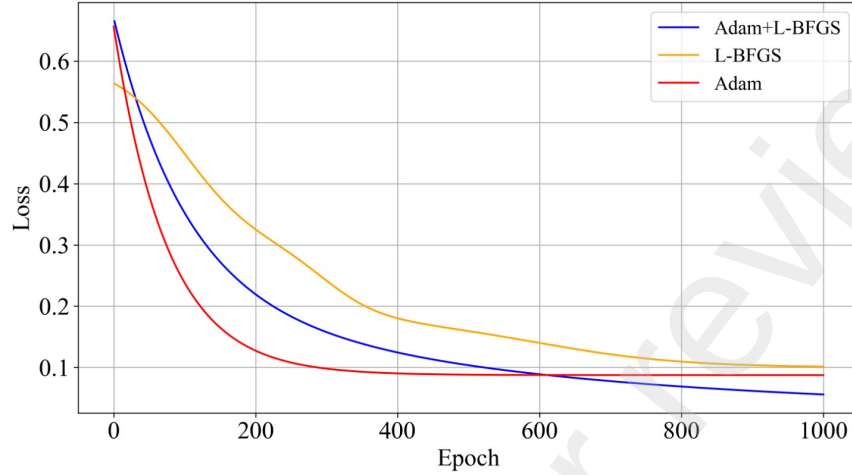


Fig. 12. Comparison of loss convergence under different optimization strategies.

(3) Without hybrid optimization:

- Adam Only: This reduces the number of iterations from the baseline 600 to 260, but incurs a substantial increase in RE, suggesting that although Adam accelerates convergence, it sacrifices predictive precision.
- L-BFGS Only: This increases iterations to 800 and likewise raises RE, showing that although second-order optimization can achieve finer parameter updates, its lack of first-order acceleration impairs both efficiency and final accuracy.

These results confirm that alternating between Adam and L-BFGS achieves an optimal tradeoff between training speed and predictive performance.

In summary, the physical constraint is fundamental to ensuring model precision and interpretability; KANs excel at modeling the intricate nonlinear relationships inherent in physics-informed data fusion; and the hybrid Adam and L-BFGS optimization strategy markedly enhances both training efficiency and convergence stability.

535 6. Conclusion

536 In this paper, we developed PIKANs, a landslide instability-time predic-
537 tion method based on GB-SAR monitoring data, which markedly improves the
538 accuracy and timeliness of early warning. The key contributions are:

539 (1) **Accurate detection of the SP.** To overcome the detection lag caused
540 by noise in conventional approaches, we introduce a dynamic SP positioning
541 strategy based on normal-distribution confidence intervals, enabling real time
542 capture of the landslide’s acceleration phase.

543 (2) **Dynamic parameter calibration and the advantage of high-**
544 **frequency monitoring.** At the Dexing open-pit mine, leveraging minute-level,
545 real-time, high-frequency, high-precision GB-SAR monitoring allows PIKANs
546 parameters to be continuously updated, ensuring predictions remain close to
547 the actual failure time. At the Xinjing open-pit mine, however, SBAS-InSAR
548 is constrained by revisit intervals (typically several days to 2 weeks), making
549 it difficult to capture short-term deformation and causing a general decline in
550 prediction accuracy across all models. Although PIKANs delivers the best per-
551 formance on both datasets, high-frequency monitoring clearly plays a decisive
552 role in precisely predicting landslide failure times.

553 (3) **Experimental validation and performance enhancement.** Using
554 field-based GB-SAR datasets from an open-pit mine, PIKANs reduces the RE
555 of instability-time prediction by 30.23% compared to a PINN baseline and by
556 24.24% compared to an improved KAN model (TimeKAN), demonstrating a
557 substantial gain in predictive accuracy. Although PIKANs delivers exceptional
558 results during the acceleration stage, predicting failure in the constant rate and
559 deceleration phases remains challenging. Future work will focus on:

- 560 • Developing generalization strategies for limited-data scenarios across di-
561 verse geological settings.
- 562 • Extending model adaptability to varied deformation patterns (e.g., constant-
563 rate, deceleration).

- Optimizing real-time early-warning performance to enhance the method's practicality and robustness.

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